

Mutual Fund Analytics: Performance, Risk & Investor Insights

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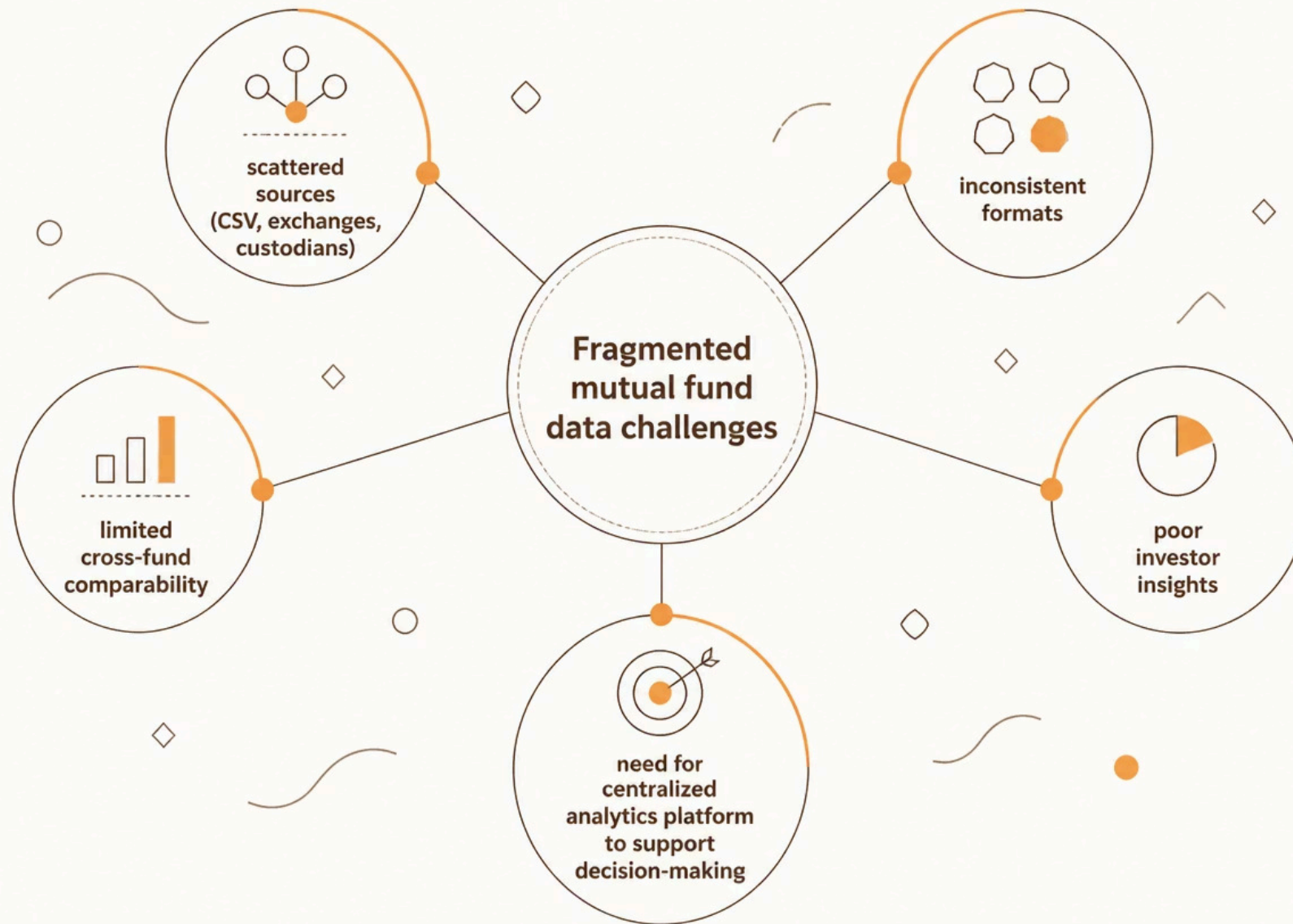
Capstone project demonstrating end-to-end analytics: data engineering, exploratory analysis, risk modelling, and interactive dashboards.

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Problem Statement



Current gaps: data is dispersed and inconsistent, investors cannot easily compare risk-adjusted returns, and organisations lack a unified analytics platform to generate actionable insights.

Project Objectives



Build ETL pipeline

Automate ingestion from NAV, AUM, SIP and holdings sources.



Clean & structure data

Standardise timestamps, handle missing NAVs, normalise categories.



SQL database design

Relational schema (funds, fact_nav, transactions, holdings).



Perform EDA

Trend analysis, seasonality, cohort behaviour.



Calculate performance metrics

CAGR, Sharpe, Sortino, Alpha/Beta, drawdowns.



Analyze risk

VaR/CVaR, rolling Sharpe, sector concentration (HHI).



Build dashboard

Interactive Power BI for fund comparison and drill-downs.



Create recommendation engine

Rule-based + score-driven fund suggestions for investors.

Data Sources

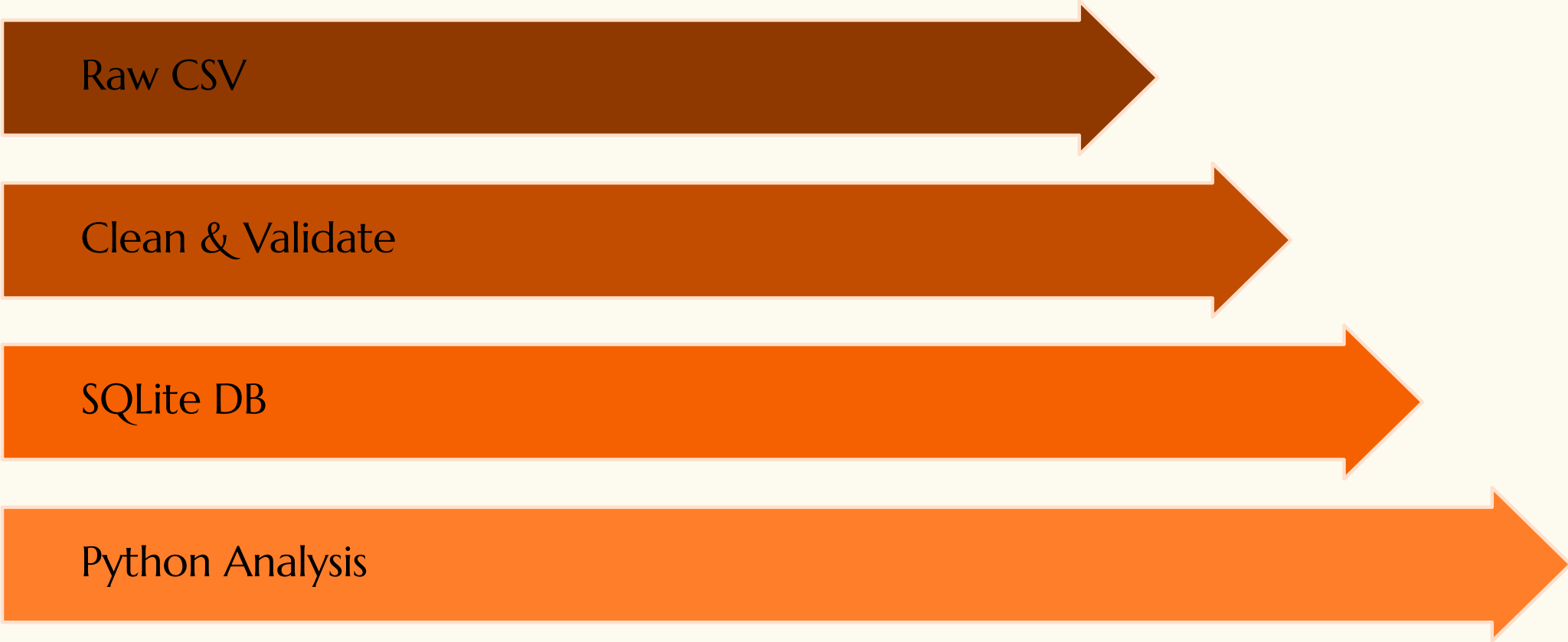


Key inputs consolidated into the analytics pipeline:

- NAV history (daily)
- AUM time series
- SIP transaction records
- Category inflows & industry flows
- Investor transactions & folio data
- Portfolio holdings (sector, weight)
- Benchmark indices (local & global)

Each source mapped with provenance, update cadence, and quality checks.

System Architecture



Raw CSV

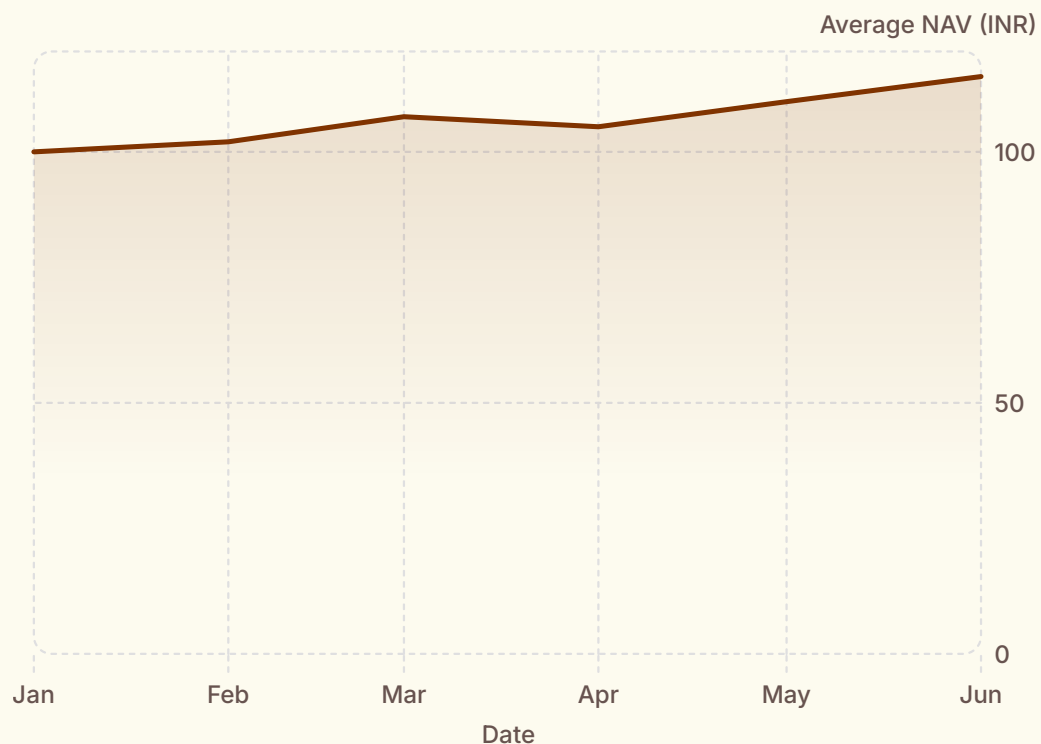
Clean & Validate

SQLite DB

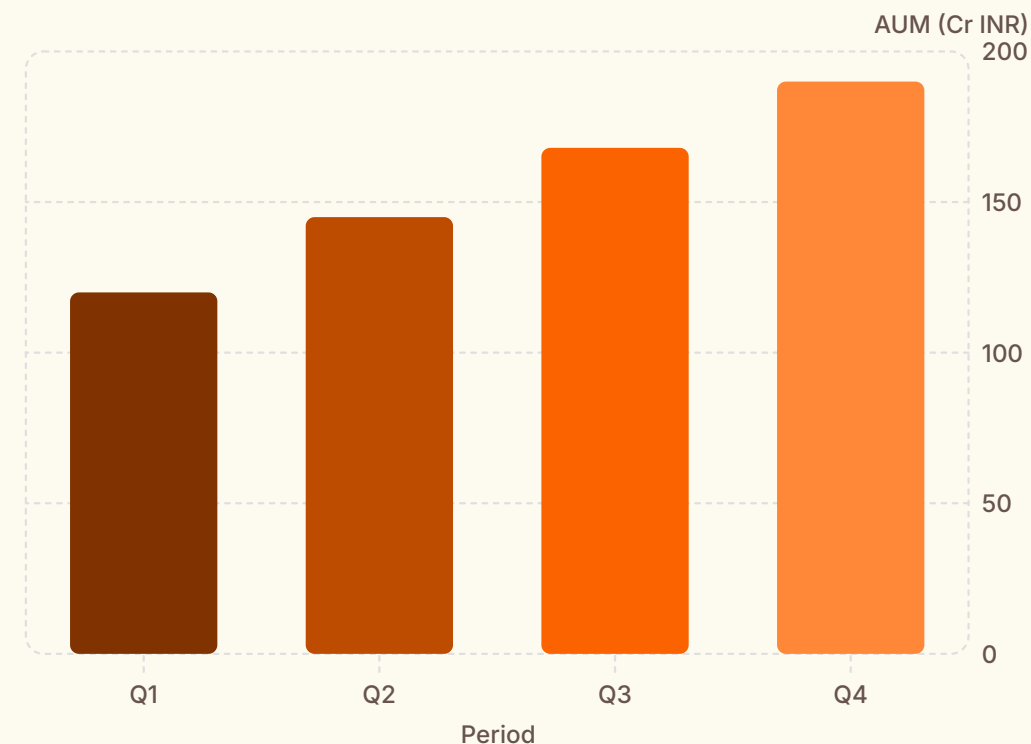
Python Analysis

Lightweight, reproducible architecture for internship-grade analytics: reproducible ETL, single-file SQLite for portability, Python for modelling, Power BI for delivery.

EDA Highlights — Trends

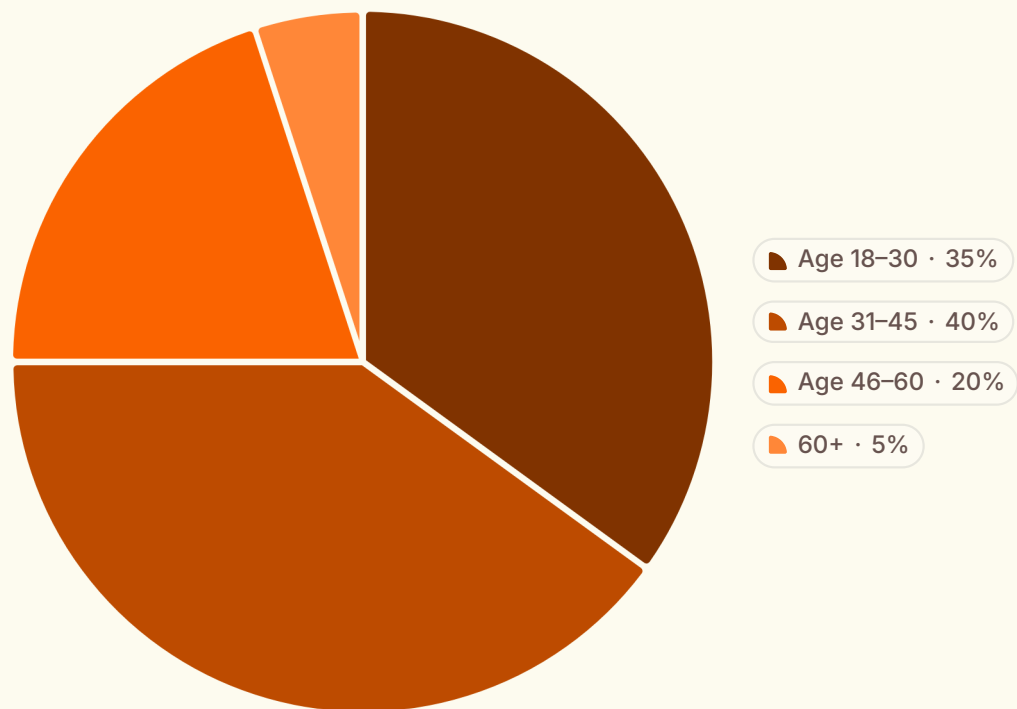


NAV trends reveal growth punctuated by market drawdowns — useful for return and volatility decomposition.



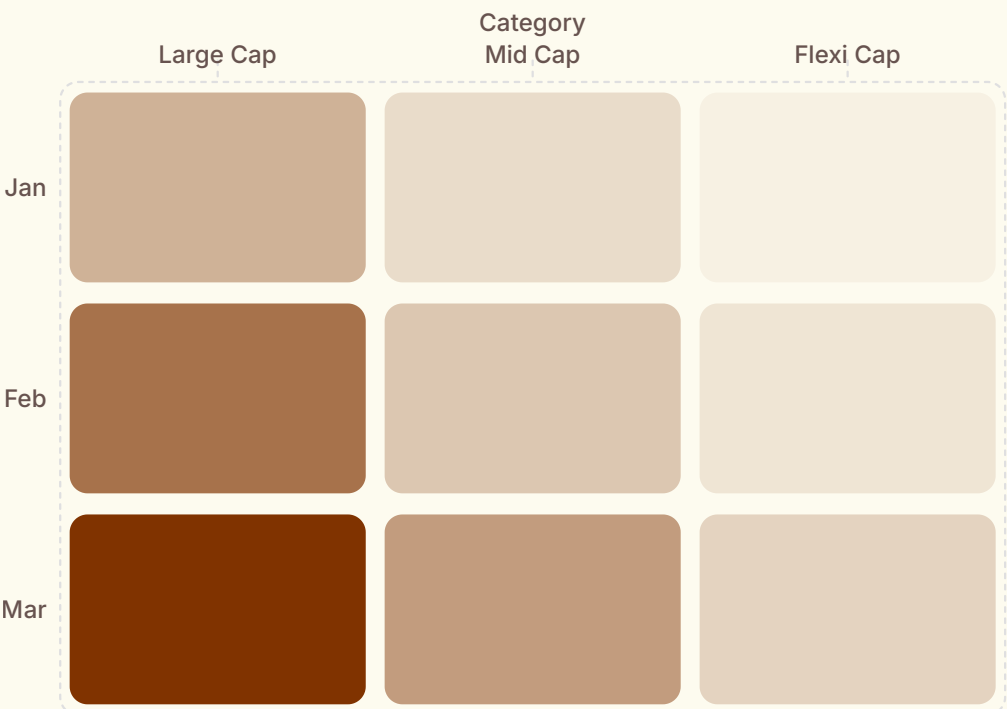
AUM growth indicates sustained inflows; combine with NAV trend to separate market returns from net new money.

EDA Highlights — Investor & Category Insights



Investor demographics skew younger — implications for product design and marketing.

Geographic mapping and folio growth charts (interactive) used to locate city-level demand—T30 cities drive majority of SIP volumes.



Category inflow heatmap highlights where retail money is moving month-to-month.

Performance Analytics

CAGR

Annualised return over chosen horizon; baseline for investor expectations.

Sharpe & Sortino

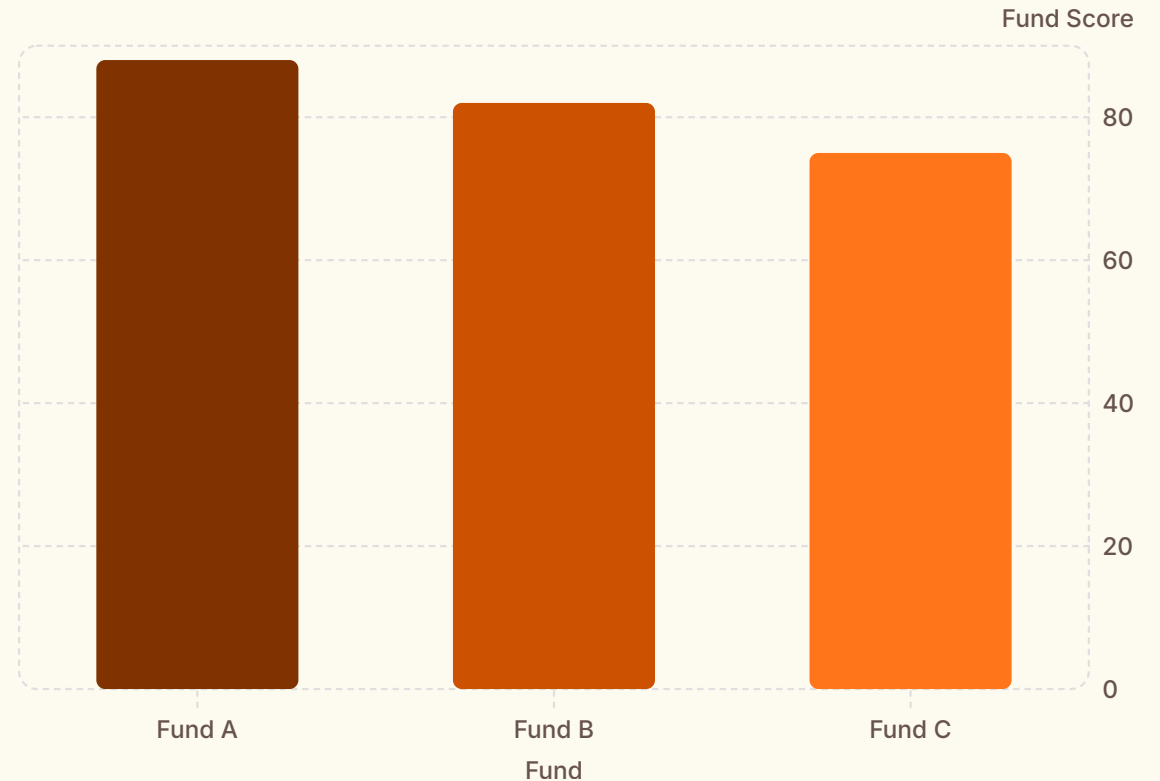
Risk-adjusted returns — Sharpe for total volatility, Sortino for downside focus.

Alpha & Beta

Manager skill vs systematic exposure relative to benchmarks.

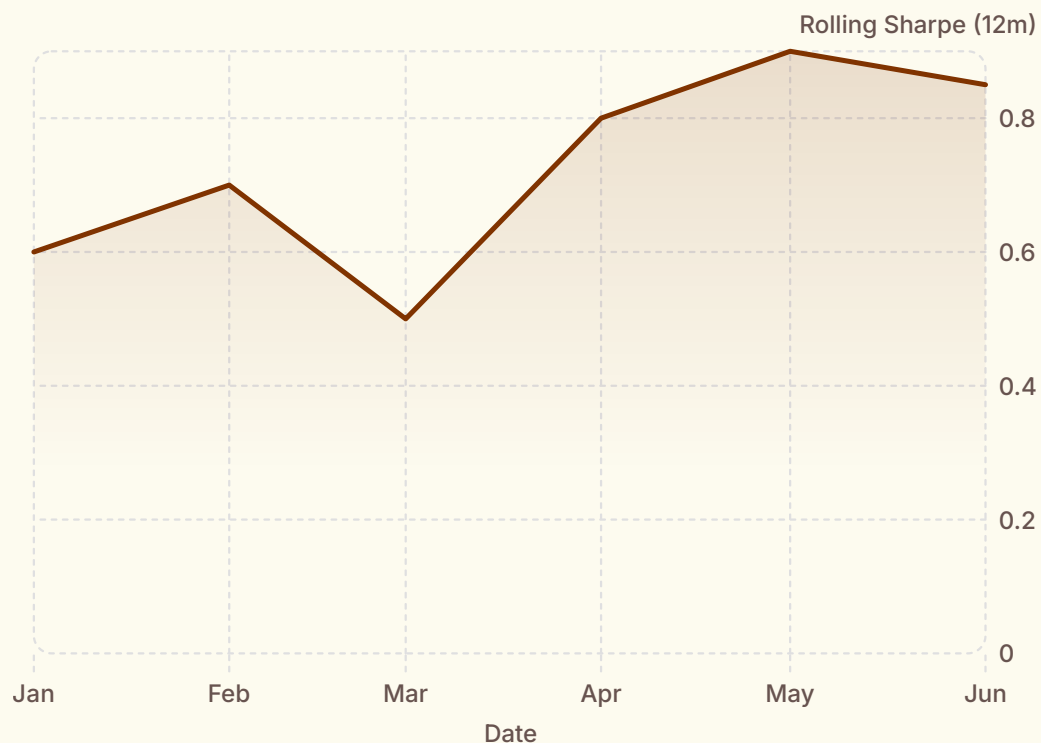
Max Drawdown & Scorecard

Downside risk and composite fund scoring for ranking.



Composite score blends performance and risk metrics to produce intuitive rankings for mentors and evaluators.

Risk Analytics



Rolling Sharpe reveals changing risk-adjusted performance across market cycles.

1

VaR (95%)

Estimated one-day loss threshold for extreme events.

2

CVaR

Average loss beyond the VaR — tail risk exposure.

3

SIP Continuity

Analysis of investor SIP persistence and churn impact.

4

Sector HHI

Concentration measure to flag concentration-driven risk.

Dashboard & Key Deliverables

Interactive Mutual Fund Analytics Dashboard — Power BI (three pages): overview, fund comparison, investor cohorts.



Deliverables

ETL scripts, SQLite dataset, Python notebooks, Power BI reports, README with reproducible steps.

Next Steps

Deploy pipeline, add live API feeds, productionise recommendation engine.

31-03-2022

31-12-2025

TOTAL AUM (₹ Cr)

39M

Total Folios (Cr)

416.42

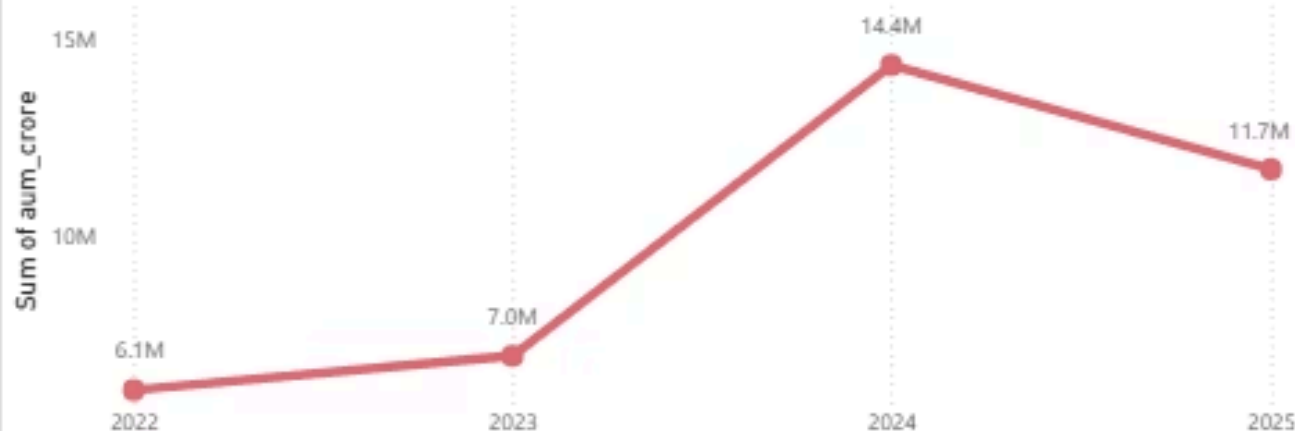
Total Schemes

40

Total SIP Inflows (₹ Cr)

940K

Industry AUM Trend



AUM by Fund House

